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A new parameter-free correlation functional based on an average atomic reduced density gradient analysis

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A new parameter-free correlation functional based on the local Ragot-Cortona approach [J. Chem. Phys. **121**, 7671 (2004)] is presented. This functional rests on a single ansatz for the gradient correction enhancement factor: it is assumed to be given by a simple analytic expression satisfying some exact conditions and containing two coefficients. These coefficients are determined without implementing the functional and without using a fitting procedure to experimental data. Their values are determined by requiring that the functional gives a correct average reduced density gradient for atoms, which, to some extent, can be considered an *intrinsic* atomic property. The correlation functional is then coupled with the Perdew-Burke-Erzernhof (PBE) exchange and compared with the original PBE approach as well as with some other pure density or hybrid approaches. Standard tests for atomic and molecular systems show that our new functional significantly improves on PBE, showing very interesting properties. © *2008 American Institute of Physics*.
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I. INTRODUCTION

Density-functional theory (DFT) is nowadays one of the widespread tools of quantum chemistry, notably thanks to its reliability, low computational cost, and range of applications, which spans from atomic calculations to supramolecular modeling and solid-state chemistry.[1] In the Kohn-Sham approach to DFT, the only contribution to the total electronic energy that must be approximated is the exchange-correlation term.[2] Finding the most appropriate approximation for the exchange-correlation functional certainly constitutes one of the greatest challenges of modern DFT development and it appears to be vital for an accurate evaluation of physicochemical properties.[3]

In order to improve existing analytical expressions for the functionals or to design new ones, two different approaches can be envisaged. The first strategy consists in assessing a certain mathematical form for the functional, with some unknown *parameters* that can be determined by fitting to a set of experimental data, the so-called "training set" (examples of such datasets can be found in Ref. 4). The functionals determined in such a way give accurate results for a large set of systems and properties, not necessarily included in the training set. The VSXC (Ref. 5) or the HCTH (Ref. 6) functionals, just to cite some of them, have been determined following this philosophy. Another way of working is requiring that the mathematical expression of the functional satisfies some theoretical conditions that are known to be verified by the exact exchange-correlation functional. Such a strategy, which does not necessarily provide the user with clear hints about the "chemical" applicability of the designed functional, leads to the determination of the *coefficients* present in the chosen form. Few functionals belong to this category and the most representative ones are those proposed by Perdew *et al.*[7] and its evolution, the Tao-Perdew-Staroverov-Scuseria[8] model. These functionals can be regarded as first principles approaches. These two categories hold also for the so-called hybrid functionals, which include a fraction of Hartree-Fock (HF) exchange. Here, the B3LYP approach[9,10] is certainly one of the best known approximations belonging to the first category, while the Perdew-Burke-Erzernhof (PBE) hybrid functional[11,12] can be considered as belonging to the second category, provided that a well-defined ansatz for the adiabatic connection is accepted.[13] Eventually, the HF contribution can be tuned in order to minimize the error on a specific properties or sys-

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tem. Of course, both approaches are not exclusive and functionals with a limited number of parameters (1 or 2) have been developed (see, for instance, Refs. 14 and 15).

It is common practice to split any functional into an exchange and a correlation contributions. The last one, which represents about only 10% of the total exchange-correlation energy, has been the subject of a relatively small number of studies in comparison with those dealing with the exchange part. Among the most popular correlation functionals, we mention those developed by Perdew *et al.* (such as PBE or PW91)[7,16] and the one derived by Lee *et al.*,[17] which stems from the Colle-Salvetti (CS) analysis.[18] Few years ago, Krieger *et al.*[19] have proposed a new meta–generalized gradient approximation (meta-GGA) functional, based on the idea of a uniform electron gas with a gap in the excitation spectrum, leading to competitive results.[20] All these functionals have the characteristics of having few [Lee-Yang-Parr (LYP), PW91] or no (PBE, KCIS) parameters.

More recently, Ragot and Cortona have developed a local correlation functional[21] characterized by a very simple analytical expression. This functional, correcting some of the Colle-Salvetti original inconsistencies, does not contain any adjusted parameter and it can be considered as a significant improvement with respect to the traditional local-density approximation schemes.[22,23] Indeed, the performance gap with respect to the gradient corrected approximations (GGAs) is strongly reduced, if not vanished, for some physicochemical properties (e.g., atomization energies and ionization potentials).[22,23]

Starting from this local approach, we have thought interesting to develop a new parameter-free GGA correlation functional. The interest in such approach resides, beyond the intrinsic theoretical interest, in some computational aspects. In fact, even if hybrid methods have significantly improved the numerical reliability of DFT approaches and can be considered as the first choice between all the available functionals, pure exchange-correlation functionals have still a predominant role in many chemical applications. For example, there is an increasing interest in the *ab initio* molecular dynamics approach, where the use of plane waves precludes a fast evaluation of the HF exchange. In a similar manner, many solid-state codes use only pure DFT approaches. All these cases call for a reliable and efficient treatment of correlation (and exchange) energies, without the inclusion of the exact exchange.

II. THE DETERMINATION OF THE GRADIENT CORRECTED FUNCTIONAL

A. The general factorized form for the correlation functional

In the GGA approach, the correlation energy density can be expressed as a function of three variables,[7]

$$E_c^{\mathrm{GGA}} = \int e_c(r_s(\mathbf{r}), s(\mathbf{r}), \zeta(\mathbf{r})) d^3 r, \tag{1}$$

where r_s is the Seitz radius

$$r_s = \left(\frac{3}{4\pi\rho}\right)^{1/3}, \tag{2}$$

s is the so-called reduced density gradient

$$s = \left(\frac{3}{2\pi}\right)^{1/3} \|\nabla r_s\| = \frac{\|\nabla\rho\|}{2(3\pi^2)^{1/3}\rho^{4/3}}, \tag{3}$$

and ζ is the spin-polarization variable defined as

$$\zeta = \frac{\rho_\uparrow - \rho_\downarrow}{\rho}. \tag{4}$$

A convenient simplification is narrowing the set of the possible functions by imposing a separation of the variables, that is to say, by choosing the following factorized form:

$$E_c^{\mathrm{GGA}} = \int A(r_s(\mathbf{r})) B(s(\mathbf{r})) C(\zeta(\mathbf{r})) d^3 r. \tag{5}$$

Here, A is the local contribution, B the gradient correction, and C the spin-polarization factor. All these terms can be determined separately. So the Ragot-Cortona (RC) functional can be chosen for the local contribution,

$$A(r_s(\mathbf{r})) = \frac{3}{4\pi r_s^3}\varepsilon_c^{\mathrm{RC}}(r_s(\mathbf{r})). \tag{6}$$

For the spin-polarization term, a simple function has been proposed by Wang and Perdew,[24]

$$C(\zeta(\mathbf{r})) = \left(\tfrac{1}{2}[(1+\zeta(\mathbf{r}))^{2/3} + (1-\zeta(\mathbf{r}))^{2/3}]\right)^3. \tag{7}$$

The last term B remains to be determined.

B. The local contribution: The Ragot-Cortona functional

The RC correlation functional is given by the following expression:[21]

$$\varepsilon_c^{\mathrm{RC}} = \frac{-0.655\,868 \arctan(4.888\,270 + 3.177\,037 r_s) + 0.897\,889}{r_s}, \tag{8}$$

where ε_c is the correlation energy per electron. The starting point of Ragot and Cortona was a CS-like correlated wave function, given by a product of the HF wave function times a correlation factor,

$$\psi(x_1, \ldots, x_N) = \psi^{\mathrm{HF}}(x_1, \ldots, x_N) \prod_{i<j} [1 - f(\mathbf{r}_i, \mathbf{r}_j)], \tag{9}$$

where x_i indicates the space and spin coordinates of the electron i, and the correlation function f is given by

$$f(\mathbf{r}_1, \mathbf{r}_2) = \left[1 - \Phi(\mathbf{R}_{12})\left(1 + \frac{r_{12}}{2}\right)\right] e^{-\beta_c^2(\mathbf{R}_{12}) r_{12}^2}. \tag{10}$$

Here, $\mathbf{R}_{12}$ and r_{12} are the center-of-mass and the relative coordinates of the electrons 1 and 2, respectively,

$$\mathbf{R}_{12} = (\mathbf{r}_1 + \mathbf{r}_2)/2, \quad r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|. \tag{11}$$

The Gaussian function appearing in the f expression determines the range of the correlation interactions which are taken into account. This range is fixed by β_c, which can be roughly interpreted as the reciprocal of a correlation hole radius. CS assumed that β_c is related to the electron density as follows:

$$\beta_c(\mathbf{R}_{12}) = q[\rho(\mathbf{R}_{12})]^{1/3}, \tag{12}$$

where q is a parameter and, after some theoretical developments, they wrote the function Φ in terms of β_c as follows:

$$\Phi = \frac{\sqrt{\pi}\beta_c}{(1 + \sqrt{\pi}\beta_c)}. \tag{13}$$

In order to derive the RC correlation energy expression, an approximate one-electron reduced density matrix was determined from the wave function given in Eq. (9). Then, applying the resulting expression to the homogeneous electron gas, Ragot and Cortona were able to obtain the kinetic correlation energy per electron,

$$t_c(r_s) = \frac{1}{11.9475 + 14.9062 r_s + 4.8440 r_s^2}. \tag{14}$$

Equation (8) was then deduced inverting the well-known relation,[25]

$$t_c = -\frac{\partial(r_s \varepsilon_c)}{\partial r_s}. \tag{15}$$

Two remarks are in order. The first one is that the correlation functional given in Eq. (8) is a very simple local functional that can be easily implemented in any DFT calculation program. The second one is that the functional is parameter-free. The constants which appear in Eq. (8) [or, equivalently, in Eq. (14)] were determined without using any fitting procedure. The value of the only parameter which appears in the CS wave function [q, in Eq. (12)] was fixed by using two limit conditions. Thus, the functional given in Eq. (8) is parameter-free exactly in the same sense as the usual local-density approximation[3,26,27] is parameter-free.

C. The GGA contribution

At this stage, only $B(s)$ remains undetermined. This enhancement factor should recover the uniform electron gas (that is to say, in our approach, the polarized RC functional) when the gradient equals to 0. At the same time, in the rapidly varying limit ($s \to \infty$), the correlation must vanish.[7] These two constraints impose

$$B(0) = 1,$$

$$\lim_{s\to\infty} B(s) = 0. \tag{16}$$

A very simple ansatz respecting both relations is

$$B(s(\mathbf{r})) = \frac{1}{1 + \sigma s(\mathbf{r})^{\alpha}}. \tag{17}$$

The starting point to determine the two constants of this last equation, σ and α, is the approach developed by Zupan *et al.*[28,29] These authors showed that (for light atoms) the total exchange-correlation energy can be estimated from the knowledge of the average values of r_s, ζ, and s. They also gave suitable definitions of these average values.

Let us start, in analogy to their work, by introducing the following definition of the average value of $B(s)$:

$$\langle B \rangle = \frac{E_c^{\mathrm{GGA}}}{\displaystyle\int A(r_s(\mathbf{r}))C(\zeta(\mathbf{r}))d^3r}. \tag{18}$$

In order to compute the right side of this equation, the numerator can be replaced by the exact atomic correlation energies, provided that the GGA correlation energies are very close to the exact values. The evaluation of the denominator requires the knowledge of the electron densities arising from the GGA calculation and so, requires the knowledge of B. However, in first approximation, these GGA densities can be replaced by those derived by a spin-polarized local-density calculation, so that the integral in Eq. (18) is easily computed since it corresponds to the (self-consistent) local-density total energy. Therefore,

$$\langle B \rangle \approx \frac{E_c^{\mathrm{exact}}}{E_c^{\mathrm{local}}}. \tag{19}$$

Nevertheless, this is not sufficient to determine the parameters σ and α and a further approximation is required. According to Zupan *et al.*, it can be then assumed that

$$\langle B(s) \rangle \approx B(\langle s \rangle). \tag{20}$$

So, combining Eq. (17), (19), and (20), we obtain

$$\sigma\langle s \rangle^{\alpha} \approx \frac{E_c^{\mathrm{local}}}{E_c^{\mathrm{exact}}} - 1. \tag{21}$$

Equation (21) will allow us to determine σ and α, if $\langle s \rangle$ and E_c^{local} are known for some reference systems. In our case, the spin-polarized RC local functional [terms A and C of Eq. (5)] has been used in order to evaluate E_c^{local}, and atoms have been chosen as reference systems, since their exchange and correlation energies are known with an excellent accuracy.

TABLE I. Calculated values of $\langle s\rangle^{\text{eff}}$ (in a.u.) for He, Be, and Ar, using six different densities.

ρ_D	HF	SVWN	PBE	PBE0	MP4(SDQ)	CCSD
			He			
$\langle s\rangle^{\text{eff}}_{\text{PBE}}$	0.9031	0.9078	0.9083	0.9065	0.9047	0.9047
$\langle s\rangle^{\text{eff}}_{\text{PW91}}$	0.9032	0.9078	0.9084	0.9066	0.9049	0.9049
			Be			
$\langle s\rangle^{\text{eff}}_{\text{PBE}}$	0.8780	0.8773	0.8776	0.8775	0.8769	0.8769
$\langle s\rangle^{\text{eff}}_{\text{PW91}}$	0.8787	0.8779	0.8782	0.8781	0.8774	0.8772
			Ar			
$\langle s\rangle^{\text{eff}}_{\text{PBE}}$	0.6209	0.6203	0.6208	0.6207	0.6206	0.6206
$\langle s\rangle^{\text{eff}}_{\text{PW91}}$	0.6160	0.6155	0.6159	0.6159	0.6159	0.6159

Furthermore, an average value $\langle s\rangle$ can be defined for any atomic or molecular systems in terms of the exchange energy only.[29] The starting point is a GGA exchange functional written as follows:

$$E_x^{\text{GGA}} = \int e_x^{\text{uniform}}(r_s(\mathbf{r}))F^{\text{GGA}_x}(s(\mathbf{r}))d^3r. \tag{22}$$

where F^{GGA_x} is the so-called enhancement factor. Given a density ρ_D, the following ratio can be calculated:

$$R^{\text{GGA}_x}[\rho_D] = \frac{E_x^{\text{GGA}}[\rho_D]}{E_x^{\text{uniform}}[\rho_D]}. \tag{23}$$

A simple and convenient average value of s, $\langle s\rangle^{\text{eff}}$, is found by solving the following equation:

$$F^{\text{GGA}_x}(\langle s\rangle^{\text{eff}}) = R^{\text{GGA}_x}[\rho_D]. \tag{24}$$

This equation can be solved, analytically or numerically, for any exchange functional having an analytical form, such as PBE or PW91. *A priori*, $\langle s\rangle^{\text{eff}}$ depends on the choice of GGA_x and of ρ_D. It is clear, however, that $\langle s\rangle^{\text{eff}}$ can be an useful quantity only if its value is relatively independent of these choices. In order to check this point, $\langle s\rangle^{\text{eff}}$ has been calculated using the PBE and PW91 exchanges, and several densities, obtained using different DFT, HF, and post-HF methods for three unpolarized atoms, namely, He, Be, and Ar. It must be noticed that post-HF densities had been used to compute correlation potentials.[30]

The results are presented in Table I. The computed $\langle s\rangle^{\text{eff}}$ values for He, Be, and Ar are 0.91, 0.88, and 0.62 a.u., respectively. Furthermore the variations of $\langle s\rangle^{\text{eff}}$ for the same atom obtained with different ρ_D or GGA enhancement factors are smaller than 0.01 a.u. It is natural to believe that the results would be identical (at the mentioned level of accuracy) if the exact densities of these atoms had been used. Accordingly, $\langle s\rangle^{\text{eff}}$ can be considered an "intrinsic" property of each atom.

In Fig. 1, $\ln(\langle s\rangle^{\text{eff}})$ is reported as a function of the Neperian logarithm of the atomic number. A linear regression enables us to write

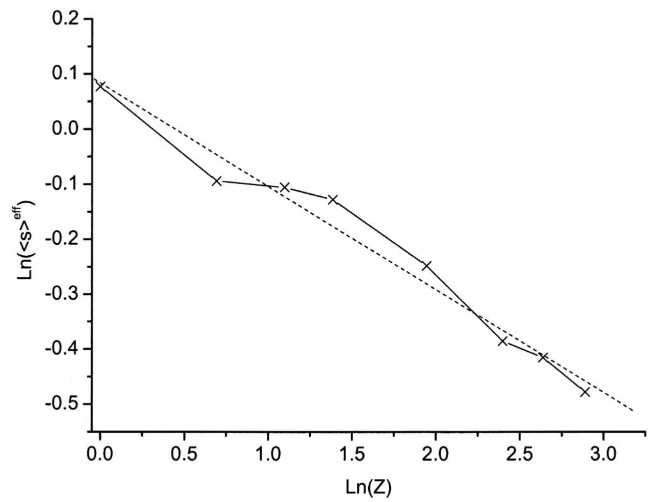

FIG. 1. Evolution of $\ln(\langle s\rangle^{\text{eff}})$ as a function of $\ln(Z)$ (full line) and the corresponding linear regression (dashed line) (see text for details).

$$\langle s\rangle^{\text{eff}} = \frac{A}{Z^{\beta}}, \tag{25}$$

where $A=1.087$, $\beta=0.187$, and Z is, as usual, the nuclear charge.

It must be pointed out that another average value for s can be determined from the distribution function $g_3(s)$ defined as[28,29]

$$g_3(s) = \int \rho(\mathbf{r})\delta(s-s(\mathbf{r}))d^3r. \tag{26}$$

This distribution is normalized to the number of electrons N and counts how many electrons correspond to a value of s between s and $s+ds$. Using g_3, another natural definition of the average value of s (referred to as $\langle s\rangle^{\text{den}}$) is

$$\langle s\rangle^{\text{den}} = \frac{1}{N}\int_0^{\infty} s g_3(s)ds. \tag{27}$$

A numerical integration of Eq. (27) for the argon atom leads to $\langle s\rangle^{\text{den}}=0.65$ a.u., a value very close to the corresponding value of $\langle s\rangle^{\text{eff}}$ (0.62 a.u.). Similar results have been obtained for other atoms, thus well underlining the "fundamental" character of $\langle s\rangle$, independent of the evaluating procedure.

For given values of σ and α, it is now possible to determine the left term of Eq. (21) for any atom. The right term is simply evaluated by calculating the self-consistent correlation energy with the given local functional (e.g., the polarized RC) and dividing the result by the exact correlation value. Figure 2 represents $\ln(m-1)$, where m is the ratio $E_c^{\text{local}}/E_c^{\text{exact}}$ with respect to the logarithm of the atomic number. A linear regression allows us to establish the relation

$$m-1 = BZ^{\gamma}, \tag{28}$$

where $B=1.731$ and $\gamma=-0.430$. The determination of α and σ is achieved by using Eqs. (21), (25), and (28). The result is

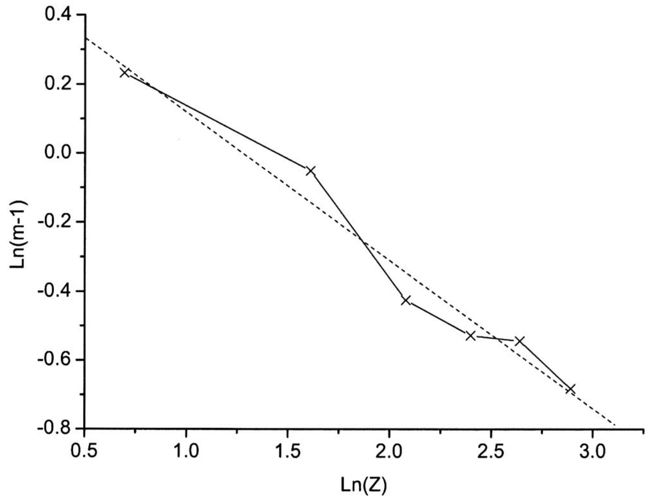

FIG. 2. Evolution of $\ln(m-1)$ as a function of $\ln(Z)$ (full line) and the corresponding linear regression (dashed line).

$$\alpha = -\frac{\gamma}{\beta} = 2.30,$$
$$\sigma = \frac{B}{A^{\alpha}} = 1.43. \tag{29}$$

These values have been determined without implementing the gradient correction and without fitting any data set. They also does not depend on the atom sets used for the linear regressions, since different atoms lead to a variation in the α and σ values of less than 0.02 units. This small difference does not affect the numerical performance of the resulting functional. In fact, the difference in the computed atomic correlation energies is less than 10^{-3} a.u. and the mean absolute errors for the atomization energies of the G2 sets are similar at more than 0.05 kcal/mol. Thus, it can be derived that σ and α show a negligible dependency on the initial choice of the atoms. Adopting the σ and α values given in Eq. (29), the final form of the gradient corrected correlation functional is established.

D. The exchange counterpart

In order to perform Kohn-Sham calculations, the new GGA correlation functional must be combined with an exchange counterpart. As the proposed functional respects the same limits of the PBE approach, the PBE exchange is a natural choice. This exchange is given by an expression containing two parameters: κ, which is unequivocally fixed by the local Lieb-Oxford limit,[31] and μ, determined so that the local-density approximation is recovered for the linear response of the uniform electron gas. In other terms, μ is chosen in such a way that the second order terms in the developments (with respect to s) of the exchange and correlation energy densities exactly compensate. Since the PBE exchange energy density, for small s values, is given by

$$e_x^{\mathrm{PBE}} = -A_x\rho(\mathbf{r})^{4/3}(1 + \mu s(\mathbf{r})^2) \tag{30}$$

and the α constant, in our functional, is different from 2, it is clear that it will be impossible (whatever the value of μ may be) that the gradient corrections to the correlation and the exchange energies compensate each other in this limiting case. However, if the residual gradient correction cannot vanish, it can be sufficiently close to 0 so that the exchange-correlation functional has the correct behavior. In the limit of low density ($r_s \to \infty$) the unpolarized Ragot-Cortona functional has the following behavior:

$$e_c^{\mathrm{RC}} = \rho(\mathbf{r})\varepsilon_c^{\mathrm{RC}}(\rho(\mathbf{r})) \approx -b\rho(\mathbf{r})^{4/3} \quad \text{with } b \approx 0.213\,341. \tag{31}$$

Furthermore, for small s,

$$B(s(\mathbf{r})) \approx 1 - \sigma s^{\alpha}. \tag{32}$$

Thus, in this regime, the gradient correction to the exchange correlation is

$$G_{\mathrm{xc}} \approx -A_x\mu\rho(\mathbf{r})^{4/3}s^2\left[1 - \frac{b\sigma}{A_x\mu}s^{\alpha-2}\right]. \tag{33}$$

The corresponding term in the original PBE approach is exactly zero. A less restrictive condition could be requiring that the expression in the brackets in Eq. (33) is very small (in absolute value), comparing to 1. Hence (if the original value of μ is conserved),

$$s < \left(\frac{2A_x\mu}{b\sigma}\right)^{1/(\alpha-2)} \approx 1.31. \tag{34}$$

This value is greater than the expected values of s for typical covalent bonds [between 0.7 and 0.9 a.u. (Ref. 32)], so that the inequality in Eq. (34) is surely verified for common chemical systems. In order to support this semiquantitative argument, we have made some tests on the atomization energies of the reduced G2 set with different values of μ. The lowest deviation has been recovered with the original μ value.

III. COMPUTATIONAL DETAILS

The resulting functional, hereafter defined with the acronym TCA, has been added to those already available in the development version of the GAUSSIAN package.[33] Several basis sets have been used in the present calculations. Atomic calculations and atomization energies have been computed using the 6-311+G(3df,2p) basis set, while geometry optimizations have been carried out at the 6-311G(d,p) level, with the exception of the metallic complexes. In this last case, the transition metal atoms have been described using the Wachters basis including a f-type polarization function,[34] while a 6-31G(d) basis has been considered for the bound ligands, following the approach of Bühl and Kabrede.[35] For the elementary transformations belonging to B6H of Lynch and Truhlar[36] and NHTBH38/04 (Ref. 37) databases, the 6-311+G(3df,2p) basis set was also used. For the proton transfer (PT) reactions, a double-ζ quality Pople basis set has been used, with polarization functions on every atoms and with diffuse functions on all the non-H atoms, namely, 6-31+G(2d,p). As recommended in Ref. 38, the hydrogen-bonded complexes have been computed using the large aug-cc-pVQZ basis without including basis set superposition error corrections in the dissociation energies values.

TABLE II. Correlation energies and relative variations of atoms belonging to the first two rows of the Periodic Table (hartree).

Atoms	Exact[a]	TCA	PBE	KCIS	LYP
H	0.000	−0.007	−0.006	0.000	0.000
He	−0.042	−0.002	0.000	0.001	−0.002
Li	−0.045	−0.013	−0.006	−0.004	−0.008
Be	−0.094	0.005	0.009	0.008	0.000
B	−0.125	0.006	0.013	0.014	0.000
C	−0.156	0.000	0.012	0.013	−0.002
N	−0.188	−0.008	0.009	0.008	−0.004
O	−0.258	0.005	0.023	0.020	0.001
F	−0.325	0.012	0.033	0.023	0.003
Ne	−0.391	0.016	0.040	0.024	0.007
MAE(H–Ne)		0.007	0.015	0.012	0.003
Na	−0.396	−0.006	0.024	0.005	−0.013
Mg	−0.438	−0.002	0.027	0.002	−0.021
Al	−0.470	−0.008	0.024	−0.001	−0.024
Si	−0.505	−0.015	0.021	−0.006	−0.024
P	−0.540	−0.025	0.014	−0.014	−0.026
S	−0.605	−0.018	0.021	−0.009	−0.023
Cl	−0.666	−0.017	0.022	−0.013	−0.024
Ar	−0.722	−0.024	0.016	−0.023	−0.028
MAE(Na–Ar)		0.014	0.021	0.009	0.023
Total MAE		0.010	0.018	0.011	0.012

[a]Reference 42.

The activation barriers are obtained by calculating single point energies on the MP2 optimized geometries. Similarly, atomization energies were computed by single point energy calculations on the geometries used by Johnson *et al.* in their study of some thermodynamic properties (G2 set).[39] The first 32 molecules of G2 were also used for geometry optimizations (G2 32 set).[40] Standard references to the basis sets and functionals can be found in Ref. 41.

IV. RESULTS AND DISCUSSION

A. Atomic correlation energies

The evaluation of the correlation energies of the atoms belonging to the first two rows of the Periodic Table is the first, simple test for the performances of the new TCA functional. The exact values (taken from Ref. 42) as well as the results obtained with different approximations are gathered in Table II. As it can be seen, the TCA functional performs better than the other considered correlation functionals, with improvements particularly notable with respect to PBE. This finding constitutes an *a posteriori* justification of the approximations made in Sec. III, so that reasoning in terms of average values appears to be legitimate.[29] It also supports the assumption that E_c^{GGA} can be replaced by E_c^{exact} in Eq. (18).

It must also be noticed that the mean absolute error (MAE) is greater for the second row (0.014 hartree) than for the first one (0.007), but the increase of the error is less important than in the case of LYP (0.023 and 0.003, respectively). Thus, the new GGA approach provides a more balanced description of the correlation energy of the atoms belonging to these two rows.

The fact that, in contrast with LYP, heavy atoms are still well described by this gradient correction can be qualitatively justified. Ragot and Cortona noticed that their local functional gives results closer and closer to the exact values as the atomic number increases. On the other hand, heavy atoms are characterized by small $\langle s\rangle$ values [Eq. (25)]. As Ragot and Cortona describe these atoms in a satisfactory way, the factor B should not be very far from 1 for them. This is consistent with Eq. (16) that imposes $B(s\equiv 0)=1$.

The ratio $E_c^{\mathrm{local}}/E_c^{\mathrm{exact}}$ decreases in a monotonic way as Z increases or, equivalently, as s decreases, and this behavior is compatible with the monotony imposed by Eq. (16), making our separation of variables grounded. In contrast, this separation would not have been possible if the trend has been the opposite: discrepancies with respect to the reference values which increase with the atomic number. In such a case, B should have small values for small gradients, but that is not possible owing to the $B(s\equiv 0)=1$ condition.

TABLE III. Mean absolute error (MAE) (kcal/mol) and maximum absolute errors (Max) (kcal/mol) for the atomization energies of the two G2 sets.

	G2-1 (55 molecules)		G2 (148 molecules)	
	MAE	Max	MAE	Max
PBE	8.1	29.1 (F_2)	17.0	50.6 (C_2F_4)
TCA	6.7	27.7 (ClF)	9.0	33.2 (C_2F_4)
B3LYP	2.3	9.9 (HOCl)	3.1	20.0 (SiF_4)

B. Atomization energies and geometry optimizations

As a second test, we have considered the calculation of atomization energies for the standard G2-1 set, composed by 55 molecules. Despite the arbitrary choice of the molecules included in this set, this selection allows for a direct comparison with previous published data and it can be considered as a standard benchmark for a first "rough" evaluation of functional performances on chemical systems. Table III summarizes the results for this test, as well as those for the extended G2 set which includes 148 molecules. Again, TCA performs better than PBE. Furthermore, the error variation in going from the G2-1 to the G2 set is much smaller in the case of our new GGA, thus suggesting a more balanced description either for the small covalent molecules and the congested systems or aromatic cycles. It is also worth noticing that, for the G2 set, the new GGA functional reduces the performance gap between PBE and the very accurate B3LYP hybrid functional of about 58%.

Considering now the structural parameters (Table IV), the PBE and TCA results are quite similar. The difference,

TABLE IV. Mean absolute errors (and absolute maximal deviations) for the bond lengths of the G2-32 and G2-M sets.

Method	G2-32 (in Å)	G2-M (in Å)
PBE	0.013 (0.067)[a]	0.014 (0.036)[b]
TCA	0.012 (0.067)[a]	0.015 (0.040)[b]
B3LYP	0.007 (0.057)[a]	0.016 (0.043)[c]

[a]C–O in HCO.
[b]Cu–O in $Cu(acac)_2$.
[c]$Cr-C^{Ar}$ in $Cr(C_6H_6)(CO)_3$.

TABLE V. Mean absolute errors (and mean signed errors) for the activation barriers of some elementary transformations (in kcal/mol) [basis set: 6-311+G(3df,2p)]. H trans., hydrogen transfer; HA trans., heavy atom transfer; nucl. sub., nuclear substitution; Uni. and asso., unimolecular and association; and PT, proton transfer.

Method	H trans.	HA trans.	Nucl. sub.	Uni and asso.	PT
PBE	9.5 (−9.5)	15.0 (−15.0)	6.9 (−6.9)	3.4 (−3.0)	4.1 (−4.1)
TCA	8.1 (−8.1)	13.7 (−13.7)	6.6 (−6.6)	2.9 (−2.7)	3.7 (−3.7)
B3LYP	4.9 (−4.9)	8.5 (−8.5)	3.4 (−3.4)	2.0 (−1.5)	1.8 (−1.8)

0.001 Å in favor of TCA for the nonmetallic systems, is obviously not significant. It is known that the bond lengths are usually too long for GGA functionals. The fact that they are too short at the Hartree-Fock level accounts for the accuracy of the hybrid models which benefit from a compensation of errors.[43]

Table IV also collects the results for the metal-ligand bond lengths of ten metal complexes [$Co(NO)(CO)_3$, $Cr(NO)_4$, $Cr(C_6H_6)(CO)_3$, $Cu(acac)_2$, $CuCH_3$, CuCN, $Fe(C_2H_4)(CO)_4$, MnO_3F, $TiCl_4$, and VF_5], forming a set referred as G2-M, which is a part of a larger ensemble recently proposed (32 molecules).[35] This small but representative set, already used in Ref. 22, includes diamagnetic as well as paramagnetic species, metals from the right and the left of first row, and various common ligands (organic, halides, and heteroelements). The conclusion remains the same and it can be noticed that TCA and PBE both perform slightly better than the very popular B3LYP functional.

C. Activation barriers

We have chosen to verify the performances of TCA on the two benchmark databases especially designed for kinetics, namely, the so-called B6H and the NHTBH38/04 sets.[36,37] The first set contains six forward-and-reverse energy barriers for three different hydrogen-transfer (HT) reactions. The second set is composed by 38 barriers for 19 reactions, including heavy atom transfer (HAT), nucleophilic substitution (NS), and unimolecular and association (UA) reactions. Once again (Table V), TCA performs better than PBE since the MAE is 16% smaller for HT, 9% smaller for HAT, 4% for NS, and 1.5% for UA reactions.

PTs are of fundamental importance in biochemistry and in biological systems. Furthermore, these reactions indeed constitute a severe challenge for DFT methods, which usually provide very low activation energies (many examples can be found in the literature, see for instance, Refs. 44 and 45), quite far from a realistic description. Four simple but representative reactions have been retained: The PT between two molecules of water ($H_5O_2^+$ system), the one in the malonaldehyde, the double PT in the formic acid dimer, and finally, the PT in the $N_2H_7^+$ system. Table V also shows that, for these reactions, TCA performs 10% better than PBE, even if the calculated barriers remain too low.

D. Hydrogen-bonded complexes

As a final test, we have optimized the ten hydrogen-bonded complexes compiled in Ref. 46. The corresponding results are gathered in Table VI, where the reference values are those obtained by Boese and Martin at a high level of correlation wave function theory. It appears that PBE almost always underestimates the hydrogen bond lengths and overestimates the dissociation energies. On the contrary, there are about as many underestimation cases as overestimation cases in the TCA or PBE0 approach. TCA divides by almost 2 the mean absolute error in the hydrogen bond lengths with respect to PBE (0.044 Å compared with 0.070 Å) and the

TABLE VI. Deviations of hydrogen bond lengths (Å) and dissociation energies (kcal/mol) for some hydrogen-bonded complexes. The aug-cc-pVQZ basis set is used in all the cases.

	Lengths				Energies			
Complex	Reference[a]	PBE[b]	TCA	PBE0[b]	Reference[a]	PBE[b]	TCA	PBE0[b]
$(H_2O)_2$	1.954	−0.030	0.002	−0.024	4.98	0.12	−0.28	−0.01
$(H_3O^+)(H_2O)$	1.195	0.016	0.016	0.001	33.74	3.29	2.56	2.52
$(H_2O)(NH_3)$	1.978	−0.059	−0.036	−0.041	6.41	0.57	0.05	0.26
$(NH_3)_2$	2.302	−0.053	0.001	−0.034	3.13	0.01	−0.36	−0.16
$(FH)(NH_3)$	1.697	−0.070	−0.060	−0.051	12.45	1.84	1.22	1.21
$(ClH)(NH_3)$	1.793	−0.196	−0.161	−0.153	8.34	2.50	1.53	1.57
$(HF)_2$	1.823	−0.037	−0.007	−0.025	4.57	0.29	−0.04	0.13
$(HCl)_2$	2.559	−0.122	−0.032	−0.067	2.01	0.07	−0.33	−0.23
(CO)(HF)	2.072	−0.001	0.042	−0.001	1.69	−0.07	−0.20	−0.09
(OC)(HF)	2.081	−0.111	−0.085	−0.069	3.53	1.05	0.61	0.52
MSE[c]	⋯	−0.066	−0.032	−0.046		0.97	0.48	0.57
MAE[d]	⋯	0.070	0.044	0.047		0.98	0.72	0.67

[a]Reference 46.
[b]Reference 38.
[c]Mean signed errors.
[d]Mean absolute errors.

mean error in the dissociation energies (respectively, 0.48 and 0.97 kcal/mol). Besides, the gap between GGA and hybrid functionals is cancelled since the mean average values for TCA and PBE0 are very similar (respectively, 0.044 and 0.047 Å for the hydrogen bond lengths and 0.72 and 0.67 kcal/mol for the dissociation energies) and the mean signed errors are even better for TCA (0.014 Å and 0.09 kcal/mol smaller). This last point seems to suggest that there is no irreducible performance gap between GGA and hybrid functionals.

V. CONCLUSIONS

In this work, we have designed a gradient correction for the RC correlation functional. The first step at this aim has been writing the functional in a factorized form, an assumption whose efficiency relies to the properties of the original RC functional. Then, an ansatz for the reduced gradient contribution has been done. A very simple analytic expression, just containing two coefficients, was chosen in such a way that the exact results are recovered in the limiting cases of large and small gradients. As we wanted to avoid the implementation of the functional and the use of fitting procedures in order to determine the parameter values, we have supposed that the gradient contribution to the total correlation energy can be approximated by its average. This average has been evaluated by means of the average value of the gradient, an approach that we have developed somewhat in details. We have shown, in particular, that the average gradient is almost independent of the density and of the functional that has been used. Accordingly, it can be regarded as an intrinsic property of each atom. Finally, we have been able to determine the two parameters of our functionals (only by using the previously existing local functional) and to justify the "average approach."

The performances of the new functional have been verified by some traditional tests. Computed atomic correlation energies are in excellent agreement with the exact values, and all the other tests (atomization energies, geometric optimization, activation barriers, and hydrogen-bonded complexes) have given better results than those obtained by the original PBE.

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